

Metals Calculator

A Lightweight CLI Tool for Real-Time Bullion Valuation

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◆ Purpose

The metals-calculator project serves as a foundational benchmark for algorithmic and quantitative developers in the early stages of their learning. I created this project to solve a personal inefficiency: manually navigating multiple websites to calculate the USD value of my bullion holdings across gold, silver, and platinum.

By integrating the OANDA API, I automated the retrieval of live metal prices and streamlined the portfolio valuation process. More importantly, this project reinforces core programming practices through clean architectural design — separating data retrieval from user interaction and computation logic. Its modular structure mirrors the scalable, maintainable systems used in professional algorithmic trading environments.

◆ Background

Traditional online calculators for precious metals often require manual entry of spot prices, lack live data integration, or offer limited customization. With growing interest in digital tools for personal finance and asset tracking, there's a need for more flexible, real-time solutions. This project fills that gap for individual investors or aspiring quants seeking both convenience and technical rigor.

◆ Technical Overview

The metals-calculator project is composed of three main architectural components:

1. API Decryption Module (demo-for-encrypting-api.py) – This handles secure access to the OANDA API by decrypting stored credentials from the macOS Keychain. The decryption module is shared across multiple projects, making it modular and adaptable to other broker APIs with minimal changes.

2. Price Retrieval Script (get_prices.py) – This script uses the requests library to make HTTP GET requests and fetch live pricing data for gold, silver, and platinum from the OANDA API.

3. Main Calculator Script (metals_calculator.py) – This script interacts with the user via the command line, requesting metal weights and calculating total portfolio value in USD using real-time prices.

The architecture separates concerns cleanly between API interaction, security handling, and user I/O, aligning with best practices in scalable software design. The sys module is used to allow flexible cross-script imports of functions such as get_decrypted_api_key(), enabling reusability and decoupled logic.

▪ Technologies Used

- Python 3
- OANDA API – For real-time pricing of gold, silver, and platinum.
- Requests Library – For handling HTTP API calls.
- Sys Module – To enable dynamic import of decryption utilities.
- macOS Keychain – For secure storage and retrieval of encrypted API credentials.

▪ Security Measures

Security is prioritized by avoiding hard-coded API keys. Instead, encrypted credentials are stored in the macOS Keychain and decrypted at runtime. This reduces the risk of accidental exposure, especially when version-controlling the project. The OANDA Account ID, which is not considered sensitive by the provider, is left hardcoded within the script.

▪ Design Limitations and Trade-offs

The calculator was originally designed for personal use, with the following trade-offs:

- Supports grams for gold and platinum.
- Supports ounces for silver.

This decision simplified input for the specific use case it was designed to serve. While this limits general usability, it reflects an intentional focus on streamlined, goal-oriented scripting over broad modularity.

♦ How It Works

▪ Description of the Command Line Interface (CLI)

The script is executed by running `metals_calculator.py`, which serves as the main entry point. It interacts with `get_prices.py` to retrieve real-time precious metal prices (gold, silver, platinum) via the OANDA API. This data is then presented in the terminal as part of a CLI.

▪ Step-by-Step Usage

1. The user runs `metals_calculator.py`.
2. Live prices for each metal are fetched and displayed.
3. The script prompts the user to input:
 - Gold weight in grams
 - Silver weight in ounces
 - Platinum weight in grams
4. It calculates the total value of each holding in USD.
5. The total value of all metals is shown, along with a 15% resale markup estimate.

▪ Sample Output

...

Live Gold Price: 3248.0 USD per ounce

Live Silver Price: 32.381 USD per ounce

Live Platinum Price: 952.083 USD per ounce

Enter the weight of your gold in grams: 10

Enter the weight of your silver in ounces: 5

Enter the weight of your platinum in grams: 5

The current value of 10.0g of gold is: \$1044.26

The current value of 5.0 oz of silver is: \$161.91

The current value of 5.0g of platinum is: \$153.05

Total value of all metals: \$1359.21

Resell for 15% markup: \$1563.09

...

▪ Input/Output Formats and Constraints

- Input Format: Numerical only (float or integer). Strings or invalid characters are not accepted.

- Output Format: USD valuation with two decimal precision.

- Constraints:

- Gold and platinum inputs must be in grams
- Silver inputs must be in ounces

♦ Setup & Requirements

▪ Python Environment

This script requires Python 3.7 or higher. It is recommended to run the program in a virtual environment to avoid conflicts with other projects.

▪ Required Libraries

Ensure the following Python packages are installed:

- requests — for HTTP requests to the OANDA API.
- sys — for importing local modules across the project structure.

```
bash
pip install requests
```

▪ OANDA API Access

To retrieve real-time metal prices, you must have an OANDA account with API access.

➡ [Register for an OANDA developer account](#)

➡ After registration, obtain your API key and account ID. These are required for script execution.

Note: The API key should not be hardcoded. In this implementation, the API key is securely stored and retrieved using macOS Keychain.

▪ File Structure

Ensure the project maintains the following structure:

```
metals-calculator/  
├── get_prices.py  
├── metals_calculator.py  
├── README.md  
├── LICENSE  
└── .gitignore
```

◆ Conclusion

▪ Summary of Value

This project stands as a strong foundational exercise in working with real-time data via APIs, making it a valuable milestone in the progression toward more advanced algorithmic and quantitative trading systems. It demonstrates secure API handling, modular function design, cross-script interaction, and a clear real-world use case—automating the valuation of precious metal holdings.

▪ Future Development

While there are no immediate plans to expand this project further due to its divergence from the Kokoro AI roadmap, potential improvements remain. For example, enhancing input flexibility to support both grams and ounces across all metals would make the tool more user-friendly. Additionally, adding functionality to accept custom metals would improve scalability. This would require transitioning to a loop-driven design, where a dynamic list of metals is handled modularly.

▪ Potential Applications

Beyond its current functionality, this tool could evolve into a historical tracking system by appending time-stamped values to a .json file during each run. This would allow users to monitor changes in their portfolio's USD value over time—paving the way for integration with future analytical dashboards or automated financial summaries.

♦ Appendix

▪ Unit Conversions Used

- 1 troy ounce (oz) = 31.1035 grams (g)
- Silver is calculated in ounces by default.
- Gold and platinum are calculated in grams by default.
- These defaults reflect the user's personal bullion holdings and can be adjusted in future modular versions.

▪ Resale Premium Estimate

- A 15% markup is applied to the calculated spot value of holdings to approximate resale premiums for physical bullion.
- This value reflects common dealer markups for small bullion products and may vary by market conditions and brand.

▪ Data Source

- Live metal prices are retrieved via the OANDA REST V20 API.